

Task23

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Domain- AiML

Q1.

Code/Output

```
23Q1.py > ...
1 import pandas as pd
2 from sklearn.datasets import fetch_california_housing
3 from sklearn.model_selection import train_test_split
4 from sklearn.linear_model import LinearRegression
5 from sklearn.metrics import r2_score
6
7 # Load dataset
8 data = fetch_california_housing(as_frame=True)
9 X = data.data
10 y = data.target
11
12 # Split data
13 X_train, X_test, y_train, y_test = train_test_split(
14     X, y, test_size=0.2, random_state=42)
15
16 # Train model
17 model = LinearRegression()
18 model.fit(X_train, y_train)
19
20 # Prediction
21 y_pred = model.predict(X_test)
22
23 # Evaluation
24 print("R² Score:", r2_score(y_test, y_pred))
25 print("\nPredictions:")
26 print(y_pred[:10])
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Aditya\OneDrive\Desktop\Mountreach\23> & c:\Users\Aditya\AppData\Local\Programs\Python\Python314\python.exe c:/Users/Aditya/OneDrive/Desktop/Mountreach/23/23.py
R² Score: 0.57578706032451

Predictions:
[0.71912284 1.76401657 2.70965883 2.83892593 2.60465725 2.01175367
2.64550005 2.16825532 2.74074604 3.91561473]

Q2.

Code/Output

```
23Q2.py> ...
1 from sklearn.datasets import load_breast_cancer
2 from sklearn.model_selection import train_test_split
3 from sklearn.linear_model import LogisticRegression
4 from sklearn.metrics import confusion_matrix, accuracy_score
5 from sklearn.metrics import precision_score, recall_score, f1_score
6
7 # Load dataset
8 data = load_breast_cancer()
9 X = data.data
10 y = data.target
11
12 # Split
13 X_train, X_test, y_train, y_test = train_test_split(
14     X, y, test_size=0.2, random_state=42)
15
16 # Train model
17 model = LogisticRegression(max_iter=5000)
18 model.fit(X_train, y_train)
19
20 # Prediction
21 y_pred = model.predict(X_test)
22
23 # Evaluation
24 print("Confusion Matrix")
25 print(confusion_matrix(y_test, y_pred))
26 print("Accuracy:", accuracy_score(y_test, y_pred))
27 print("Precision:", precision_score(y_test, y_pred))
28 print("Recall:", recall_score(y_test, y_pred))
29 print("F1 Score:", f1_score(y_test, y_pred))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Aditya\OneDrive\Desktop\Mountreach\23> & C:\Users\Aditya\AppData\Local\Programs\Python\Python314\python.exe c:/Users/Aditya/OneDrive/Desktop/Mountreach/23/23Q2.py
Confusion Matrix
[[39  4]
 [ 1 70]]
Accuracy: 0.956140350877193
Precision: 0.9459459459459459
Recall: 0.9859154929577465
F1 Score: 0.9655172413793104
```

Q3.

Code/Output

```
23Q3.py > ...
1 from sklearn.datasets import load_breast_cancer
2 from sklearn.model_selection import train_test_split
3 from sklearn.neighbors import KNeighborsClassifier
4 from sklearn.metrics import accuracy_score
5
6 # Load dataset
7 data = load_breast_cancer()
8 X = data.data
9 y = data.target
10
11 # Split
12 X_train, X_test, y_train, y_test = train_test_split(
13     X, y, test_size=0.2, random_state=42)
14
15 # Try different k values
16 for k in [3, 5, 7]:
17     model = KNeighborsClassifier(n_neighbors=k)
18     model.fit(X_train, y_train)
19     y_pred = model.predict(X_test)
20     print("k =", k, " Accuracy =", accuracy_score(y_test, y_pred))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Aditya\OneDrive\Desktop\Mountreach\23> & C:\Users\Aditya\AppData\Local\Programs\Python\Python314\python.exe c:/Us
k = 3 Accuracy = 0.9298245614035088
k = 5 Accuracy = 0.956140350877193
k = 7 Accuracy = 0.956140350877193
PS C:\Users\Aditya\OneDrive\Desktop\Mountreach\23> 
```

Q4.

Code/Output

```

23Q4.py > ...
1  from sklearn.datasets import load_breast_cancer
2  from sklearn.model_selection import train_test_split
3  from sklearn.naive_bayes import GaussianNB
4  from sklearn.metrics import classification_report, confusion_matrix
5
6  # Load dataset
7  data = load_breast_cancer()
8  X = data.data
9  y = data.target
10
11 # Split
12 X_train, X_test, y_train, y_test = train_test_split(
13     X, y, test_size=0.2, random_state=42)
14
15 # Train model
16 model = GaussianNB()
17 model.fit(X_train, y_train)
18
19 # Prediction
20 y_pred = model.predict(X_test)
21
22 print("Confusion Matrix")
23 print(confusion_matrix(y_test, y_pred))
24
25 print("\nClassification Report")
26 print(classification_report(y_test, y_pred))

```

Confusion Matrix

```
[[40  3]
 [ 0 71]]
```

Classification Report

	precision	recall	f1-score	support
0	1.00	0.93	0.96	43
1	0.96	1.00	0.98	71
accuracy			0.97	114
macro avg	0.98	0.97	0.97	114
weighted avg	0.97	0.97	0.97	114

PS C:\Users\Aditya\OneDrive\Desktop\Mountreach\23>

Q5.

Code/Output

```

23Q5.py > ...
1 from sklearn.datasets import load_breast_cancer
2 from sklearn.model_selection import train_test_split
3 from sklearn.linear_model import LogisticRegression
4 from sklearn.neighbors import KNeighborsClassifier
5 from sklearn.naive_bayes import GaussianNB
6 from sklearn.metrics import accuracy_score, precision_score
7 from sklearn.metrics import recall_score, f1_score
8 import pandas as pd
9
10 # Load dataset
11 data = load_breast_cancer()
12 X = data.data
13 y = data.target
14
15 # Split
16 X_train, X_test, y_train, y_test = train_test_split(
17     X, y, test_size=0.2, random_state=42)
18
19 models = {
20     "Logistic Regression": LogisticRegression(max_iter=5000),
21     "KNN": KNeighborsClassifier(n_neighbors=5),
22     "Naive Bayes": GaussianNB()
23 }
24
25 result = []
26
27 for name, model in models.items():
28     model.fit(X_train, y_train)
29     y_pred = model.predict(X_test)
30
31     result.append([
32         name,
33         accuracy_score(y_test, y_pred),
34         precision_score(y_test, y_pred),
35         recall_score(y_test, y_pred),
36         f1_score(y_test, y_pred)
37     ])
38
39 df = pd.DataFrame(result,
40     columns=["Algorithm", "Accuracy",
41             "Precision", "Recall", "F1 Score"])
42
43 print(df)

```

```

PS C:\Users\Aditya\OneDrive\Desktop\Mountreach\23> & C:\Users\Aditya\AppData\Local

```

	Algorithm	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.956140	0.945946	0.985915	0.965517
1	KNN	0.956140	0.934211	1.000000	0.965986
2	Naive Bayes	0.973684	0.959459	1.000000	0.979310

```

PS C:\Users\Aditya\OneDrive\Desktop\Mountreach\23> 

```

OPTIONAL Task

```
1 import streamlit as st
2
3 st.title("Machine Learning Models")
4
5 model = st.selectbox(
6     "Choose Model",
7     ["Linear Regression", "Logistic Regression", "KNN", "Naive Bayes"]
8 )
9
10 st.write("Selected Model:", model)
11
12 if st.button("Run"):
13     st.success(model + " executed successfully!")
```

